

Overview

Part 04 —  Large Language Models · 04-llm/README.md

Module 04 — Large Language Models

Where you are: Stop 4 of 16. **Time:** ~25 hours · **Prereq:** Module 03 [03-nlp] — non-negotiable.

A large language model is the Module 03 transformer, scaled, trained on most of the public internet, and then bent into a helpful assistant by post-training. This module covers the three phases of that pipeline (pretraining → alignment → inference), the decoding knobs you actually control, and the failure modes you will spend your career designing around.

Learning objectives

1. Describe the full lifecycle: pretraining → SFT → preference optimisation (RLHF/DPO) → deployment.
2. Use **scaling laws** to reason about compute, data and parameter trade-offs — and explain what Chinchilla changed.
3. Control generation deliberately: temperature, top-k, top-p, min-p, repetition penalty, and structured decoding.
4. Choose correctly between **prompting** → **RAG** → **fine-tuning** → **continued pretraining**, and defend the choice on cost.
5. Explain quantization, LoRA/QLoRA, distillation, and speculative decoding well enough to size a deployment.
6. Name the real failure modes — hallucination, sycophancy, context rot, prompt sensitivity — and mitigate each.

Suggested path

#	Do this	File	Time
1	How an LLM is built	notes/01-how-llms-are-built.md	5h
2	Using them well	notes/02-inference-and-adaptation.md	5h
3	Sampling, felt not read	code/sampling_strategies.py	2h
4	Cost & capacity model	code/llm_cost_calculator.py	2h
5	Prompt patterns that work	code/prompt_patterns.md	2h
6	Slides	slides/	1h
7	Lab	lab/EXERCISES.md	6h
8	Papers	papers/PAPERS.md	5h

The 16 terms you must own

pretraining · SFT · RLHF · DPO · scaling law · context window · temperature · top-p · logits · perplexity · in-context learning · chain-of-thought · LoRA · quantization · speculative decoding · hallucination

Exit check

Given a use case, you can produce a defensible one-page recommendation: which model, prompted or fine-tuned, at what context length, what it will cost per 1,000 requests, what its top-3 failure modes are, and how you'd measure them.